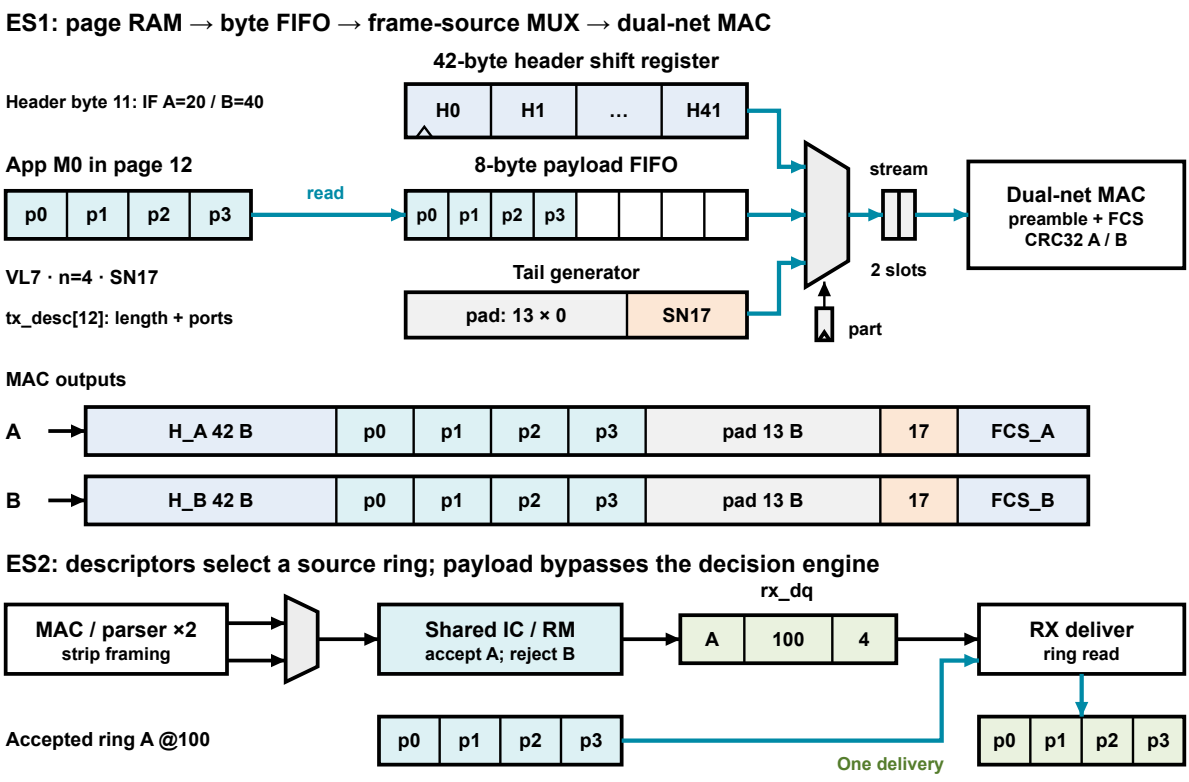
ES1: page RAM → byte FIFO → frame-source MUX → dual-net MAC
42-byte header shift register
Header byte 11: IF A=20 / B=40
H0
H1
…
H41
App M0 in page 12
8-byte payload FIFO
stream
Dual-net MAC
preamble + FCS
CRC32 A / B
p0
p1
p2
p3
read
p0
p1
p2
p3
VL7 · n=4 · SN17
Tail generator
2 slots
tx_desc[12]: length + ports
pad: 13 × 0
SN17
part
MAC outputs
A
H_A 42 B
p0
p1
p2
p3
pad 13 B
17
FCS_A
B
H_B 42 B
p0
p1
p2
p3
pad 13 B
17
FCS_B
ES2: descriptors select a source ring; payload bypasses the decision engine
rx_dq
MAC / parser ×2
strip framing
Shared IC / RM
accept A; reject B
A
100
4
RX deliver
ring read
Accepted ring A @100
p0
p1
p2
p3
p0
p1
p2
p3
One delivery